A
Mini Project Report
on

LOCKBOOK

Submitted in partial fulfilment of the requirements for the degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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Academic year: 2025-26

CERTIFICATE

This is to certify that the Mini Project report on **Secure Password Management System** has been submitted by **Kunal Kudu** (24107044), **Pranali Sawant** (24107016), **Dishti Purohit** (24107104), and **Hetvi Patel** (24107005) who are Bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfilment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Prof Kiran D Gaikwad**. Expressing gratitude towards our HOD, **Dr. Pravin Adivarekar,** and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Prof. Richa Singh and Prof Kiran Gaikwad** who gave us valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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Chapter 1

Introduction

The LockBook Secure Password Management System is designed to provide a safe and efficient solution for storing and managing multiple user credentials in a centralized platform. In today's digital era, users rely on various online services such as banking, social media, email, and e-commerce, each requiring unique login details. Managing multiple passwords becomes difficult, leading to poor practices like password reuse or storing credentials in insecure formats.

As the number of online accounts increases, users become more vulnerable to cyber threats such as hacking, phishing, and unauthorized access. Frequent password updates further increase the complexity of managing credentials. To overcome these challenges, LockBook offers a secure mobile-based solution developed using Flutter and Firebase, ensuring that user data is protected through authentication and controlled access mechanisms.

The system includes features such as Firebase Authentication, master password protection, and biometric login to enhance security. It also provides convenient functionalities like open link and copy-paste for easy access to credentials. By combining strong security measures with a simple and user-friendly interface, LockBook provides an effective solution for secure password management.

1.1 Purpose

The purpose of this document is to provide a comprehensive overview of the LockBook Secure Password Management System. It describes the design, functionality, architecture, and implementation of the system. This document is intended for students, developers, and academic evaluators who wish to understand how the system works and how it addresses the problem of secure password management.

1.2 Problem Statement

In the modern digital environment, users are required to maintain multiple accounts across various platforms, each requiring a unique and strong password. However, remembering multiple complex passwords is difficult, leading users to adopt unsafe practices such as using weak passwords, repeating passwords, or storing them in insecure locations.

These practices make users highly vulnerable to cyber-attacks such as phishing, brute-force attacks, and data breaches. Another major issue is the lack of synchronization and update tracking. When users change their passwords on external platforms, they often forget to update them in their password manager, leading to inconsistencies and access issues. Therefore, there is a need for a secure, lightweight, and user-friendly password management system that ensures data privacy, simplifies password management, and enhances overall security.

1.3 Objectives

- To design and implement a secure master authentication system that controls access to the password vault.
- To create a secure password vault that stores multiple account credentials in encrypted form.
- To enable controlled access to saved credentials through secure open links, allowing users access that app
- To enhance overall system security and maintain password data reliability.

1.4 Scope

The LockBook Secure Password Management System is designed to provide a secure and efficient solution for managing user credentials through a mobile application. The system allows users to store, organize, and manage passwords for multiple accounts in a centralized and encrypted environment.

The application is developed using Flutter for the frontend and Firebase for backend services such as authentication and database management. It supports mobile platforms including Android and requires internet connectivity for real-time data synchronization and secure access.

Future scope of the system includes enhancements such as development of a web-based version, integration of AI-based security alerts, implementation of a password strength analyser and generator, and support for multi-device synchronization. These improvements will further enhance the functionality and scalability of the system, making it more robust and adaptable for real-world applications.

Chapter 2

Proposed System

The proposed LockBook Secure Password Management System is designed to replace traditional and insecure methods of storing passwords with a secure, automated, and user-friendly solution. The system provides a centralized platform where users can safely store and manage their credentials while ensuring data privacy and protection.

The application is developed using Flutter for the user interface and Firebase for backend services such as authentication and database management. The system follows a structured approach where user data is securely stored in an encrypted format and accessed only through proper authentication mechanisms. This ensures that sensitive information remains protected from unauthorized access.

System Modules:

The LockBook system is divided into the following modules:

User Module:
This module allows users to register, log in, and manage their stored credentials. Users can add, update, delete, and view passwords securely through a simple and user-friendly interface.

Authentication Module:
This module handles secure login using Firebase Authentication. It includes master password verification and optional biometric authentication (fingerprint or face recognition) to ensure only authorized access.

Vault Module:
The vault module is responsible for storing all user credentials in encrypted form using Firebase Firestore. It ensures that sensitive data is protected and accessible only to the authenticated user.

Notification Module:

This module provides alerts related to account access and security. It helps users stay informed about login activities and improves overall system security.

2.1 Features and Functionality

The system includes the following key features:

User Features:

• Secure user registration and login
• Add, update, delete, and search stored credentials
• Open link and copy-paste functionality for easy login
• Biometric authentication support

Security Features:

• Firebase Authentication for secure login
• Encrypted storage of passwords
• Master password protection
• Controlled access to sensitive data

System Features:

• Cloud-based storage using Firebase Firestore
• Real-time data synchronization
• Scalable and reliable architecture
• Simple and responsive user interface using Flutter

Working of the System

When a user logs into the application, the authentication module verifies the user credentials using Firebase Authentication. After successful login, the system retrieves encrypted password data from the database and displays it through the user interface.

Users can perform operations such as adding new credentials, updating existing ones, or deleting unwanted entries. When accessing login details, users can use the open link and copy-paste functionality with proper permission. All operations are performed securely, ensuring that sensitive data is never exposed without authentication.

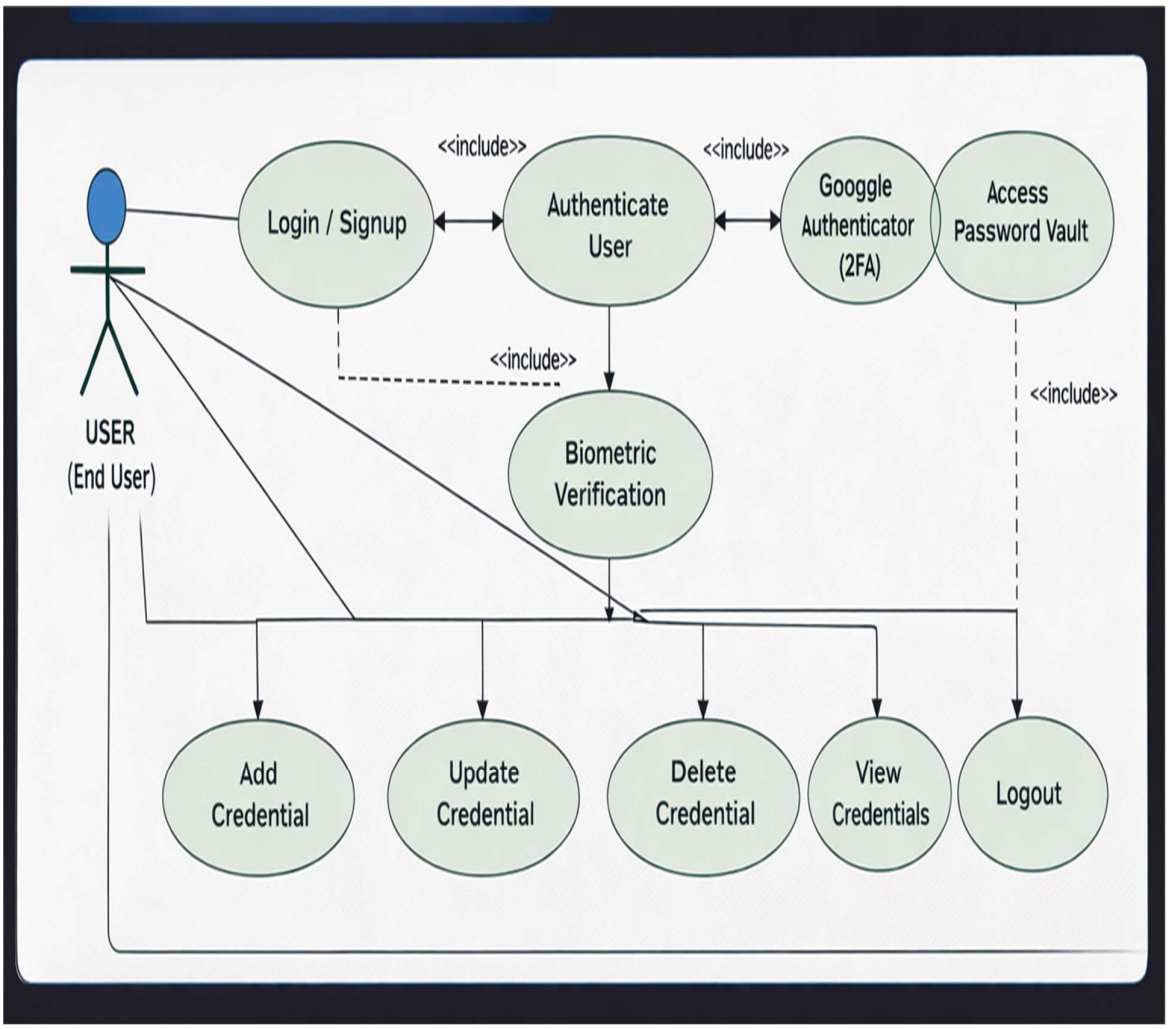

Figure 2.1: Use Case Diagram

Explanation:

The use case diagram represents the interaction between the user and the LockBook system. It shows how the user performs operations such as login, authentication, biometric verification, and Google Authenticator-based two-factor authentication before accessing the password vault. The user can manage credentials securely by adding, updating, deleting, and viewing stored data.

Chapter 3

Project Outcomes

The LockBook Secure Password Management System successfully delivers a reliable and secure solution for managing digital credentials. The system provides a centralized platform where users can store and organize multiple account passwords in an encrypted format, ensuring data privacy and protection from unauthorized access. This eliminates the need for insecure practices such as writing passwords manually or reusing the same credentials across multiple platforms, thereby improving overall digital security.

One of the key outcomes of the project is the implementation of secure user authentication using Firebase Authentication along with optional biometric verification. This ensures that only authorized users can access the password vault, thereby enhancing overall system security. The combination of authentication methods adds an extra layer of protection, making the system more robust and resistant to unauthorized access attempts.

The application offers a user-friendly interface developed using Flutter, which allows users to easily add, update, delete, and retrieve stored credentials. Features such as open link and copy-paste functionality improve user convenience by simplifying the login process across different platforms while maintaining security. The intuitive design ensures that even non-technical users can operate the system without difficulty, improving usability and user experience.

Another important outcome is the successful use of cloud-based services through Firebase Firestore, which enables real-time data synchronization and secure storage. This ensures that user data remains consistent and accessible across sessions while maintaining high performance and reliability. The use of cloud technology also allows scalability, making the system capable of handling multiple users efficiently.

Overall, the LockBook system demonstrates the practical implementation of a secure, scalable, and efficient password management solution. It highlights the importance of cybersecurity in modern applications and provides a strong foundation for further enhancements such as password strength analysis, secure sharing, and multi-device support in future versions. The project also reflects the effective integration of modern development technologies to solve real-world problems.

Chapter 4

Software Requirements

The successful development and implementation of the LockBook Secure Password Management System require a combination of appropriate hardware and software resources. These requirements ensure that the system performs efficiently, securely, and reliably across different devices and platforms.

4.1 Hardware Requirements

The hardware requirements define the minimum and recommended system specifications needed for development, testing, and execution of the application.

The system requires a processor with at least Intel Core i3 or equivalent performance to ensure smooth functioning of development tools and application execution. A minimum of 4 GB RAM is required, although 8 GB RAM is recommended for better performance, especially during development and multitasking.

The system should have at least 500 MB of free storage space for installing the application and required dependencies. A display with a resolution of 1280 × 720 or higher is recommended for better user interface visibility. Additionally, a stable internet connection is necessary for Firebase authentication, cloud data synchronization, and backup functionalities.

4.2 Software Requirements

The software requirements include the tools, technologies, and platforms used to develop and run the LockBook application.

The application is developed using Flutter, which is a cross-platform mobile development framework, and Dart as the programming language. The system uses Firebase Authentication for secure user login and registration, ensuring that only authorized users can access the application.

Firebase Firestore is used as a cloud-based NoSQL database to store user credentials securely. Encryption techniques along with packages such as flutter_secure_storage are used to protect sensitive data stored within the application.

The application supports mobile operating systems such as Android (version 8.0 and above) and iOS (version 13 and above). Development tools such as Visual Studio Code or Android Studio are used for coding, debugging, and testing the application.

4.3 System Requirements

The LockBook system is designed to meet certain functional and performance requirements to ensure efficiency and reliability.

The system should provide secure authentication and protect user data through encryption and controlled access mechanisms. It must support real-time data synchronization using Firebase services, ensuring that user data is always up-to-date.

The application should handle multiple user accounts efficiently while maintaining fast response time and smooth user experience. It should also be scalable, allowing future enhancements and feature additions without affecting system performance.

Overall, these requirements ensure that the LockBook system operates securely, efficiently, and reliably in real-world usage scenarios.

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above
RAM	4 GB minimum (8 GB recommended)
Storage	500 MB free disk space
Display	1280 x 720 resolution or higher
Internet	Required for Firebase sync and profile backup

Table 4.1: Hardware Requirements

Software Requirements

Software	Details
Operating System	Android 13.0
Programming Language	Dart
Framework	Flutter (Mobile Framework)
Database	Firebase Firestore
Encryption Library	Firebase Security Rules + flutter_secure_storage
Platform Support	Android, (via Flutter)

Table 4.2: Software Requirements

Chapter 5

Project Design

The LockBook Secure Password Management System is designed using a structured and modular approach to ensure security, scalability, and efficiency. The system architecture follows a layered design consisting of presentation, business logic, and data layers. This separation helps in maintaining clean code, easier updates, and better performance.

5.1 System Architecture

The system architecture of LockBook is based on a client-server model, where the mobile application acts as the client and Firebase services act as the backend. The architecture is divided into three main components:

• **User Interface Layer:**
This layer is developed using Flutter and provides all the screens required for user interaction such as login, dashboard, vault, and profile. It ensures a smooth and responsive user experience.

• **Application Layer:**
This layer handles the core logic of the system, including authentication, password management, and data processing. It acts as a bridge between the user interface and the database.

• **Data Layer:**

This layer is responsible for storing and retrieving data using Firebase Firestore. All user credentials are stored in encrypted form to ensure data security and privacy.

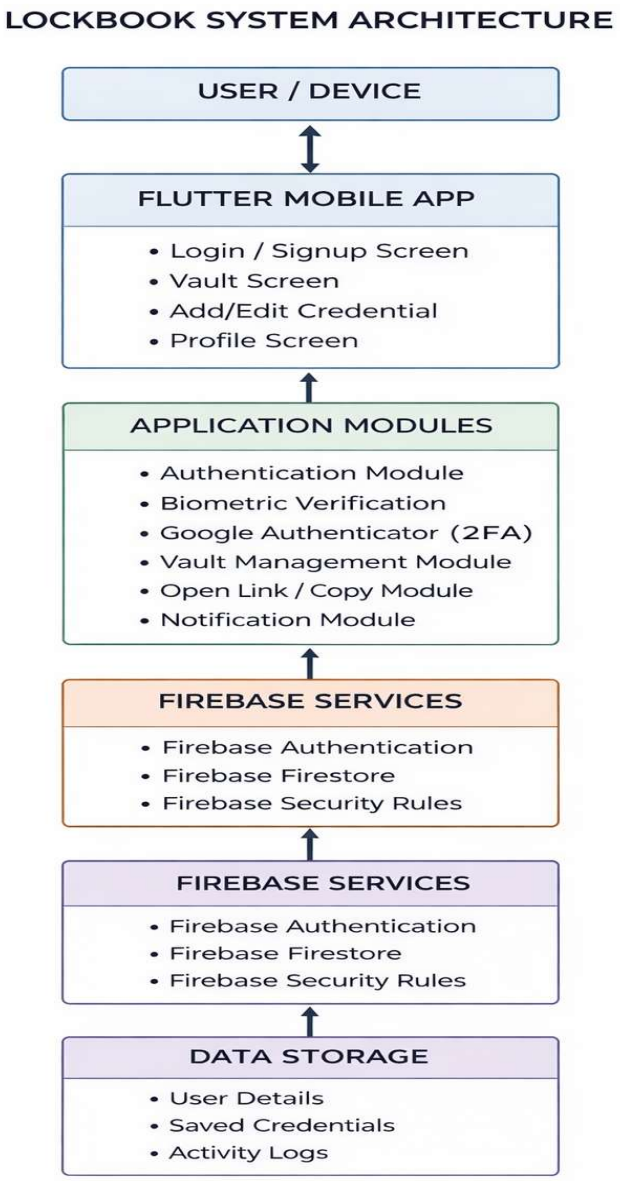

Figure 5.1: System Architecture Diagram

Explanation: The system architecture diagram shows how the user interacts with the Flutter mobile application and how the application communicates with Firebase services. It illustrates the main modules of the LockBook system, including authentication, biometric verification, vault management, and secure data storage.

Data Flow Design

The data flow in the LockBook system ensures secure communication between the user, authentication system, and database.

When a user logs into the application, the authentication module verifies the credentials using Firebase Authentication. Once verified, the system retrieves encrypted data from Firebase Firestore and displays it through the user interface.

Users can perform operations such as adding new credentials, updating existing ones, or deleting stored data. All these actions are securely processed and stored in the database.

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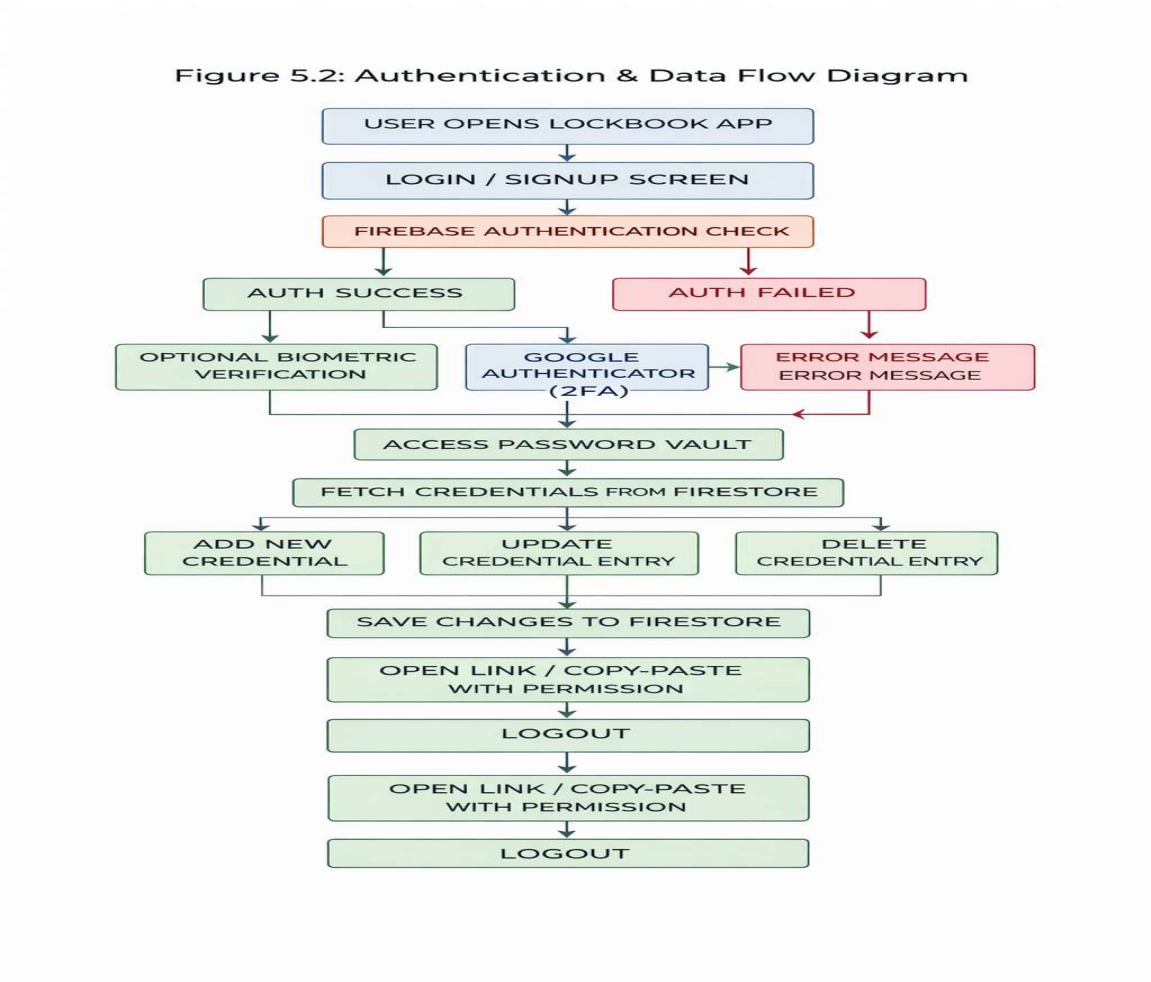


Figure 5.2: Authentication and Data Flow Diagram

Explanation:

The system also includes Google Authenticator-based two-factor authentication, which adds an additional layer of security before granting access to the password vault.

Database Design

The database design of the LockBook system is structured to store user data and credentials securely.

The database consists of the following main components:

• **User Table:**
Stores user information such as user ID, email, and authentication details.

• **Credentials Table:**
Stores website name, username, and encrypted password. All sensitive data is encrypted before storage.

• **Activity Logs:**
Stores login attempts and access records for security tracking.

All sensitive data is encrypted before being stored in the database to prevent unauthorized access. Firebase Security Rules ensure that only authenticated users can access their own data.

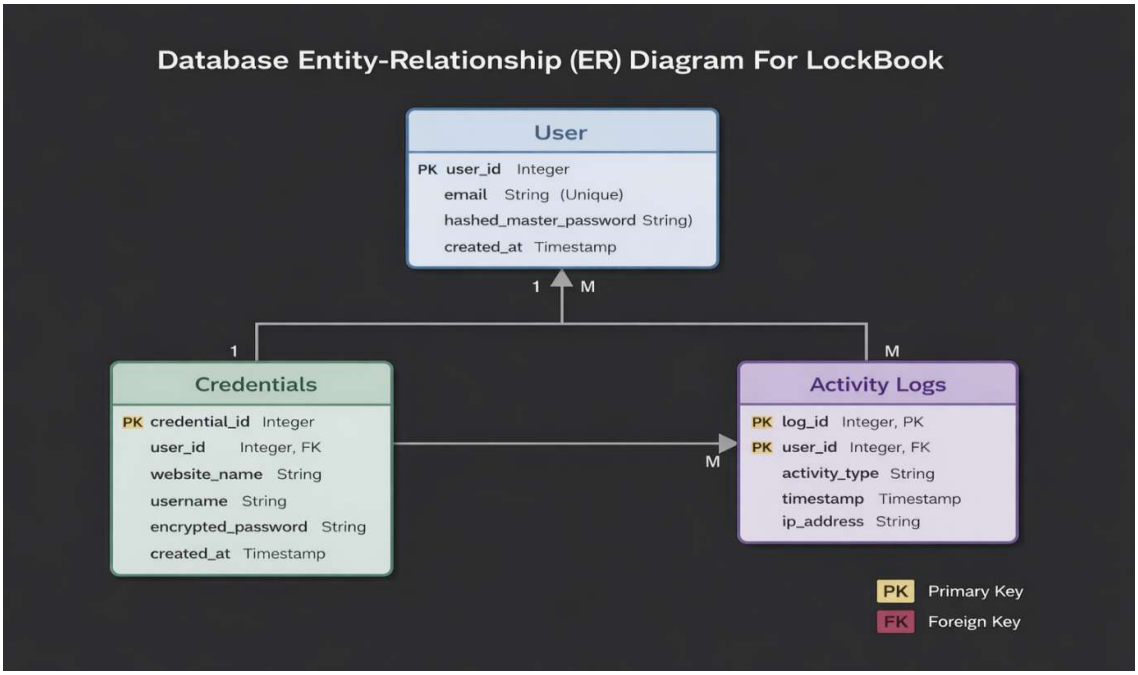


Figure 5.3: Database Entity-Relationship (ER) Diagram

Explanation:
The ER diagram illustrates the relationship between users, stored credentials, and activity logs. It shows how each user can have multiple credentials and how the system maintains secure data access and tracking.

Workflow Design

The workflow of the system describes the sequence of operations performed by the user and the system:

- User registers or logs into the application
- Authentication is verified using Firebase
- User accesses the password vault
- User can add, update, delete, or view credentials
- Data is securely stored and retrieved from the database
- User logs out securely

This workflow ensures smooth and secure functioning of the system.

Explanation:
The workflow diagram shows the step-by-step operation of the system and how users interact with different modules.

User Interface Design

The user interface of the LockBook application is designed to be simple, intuitive, and user-friendly.

The main screens include:
• Login and Signup Screen
• Dashboard/Home Screen
• Vault (Password List) Screen
• Add/Edit Credential Screen
• Profile Screen

The interface is built using Flutter and ensures smooth navigation and responsiveness across devices.

Explanation:
The UI design images represent the layout and structure of the application screens.

Overall, the project design ensures that the LockBook system is secure, scalable, and easy to use. The modular architecture and integration of Firebase services make the system efficient and reliable for real-world applications.

Chapter 6

Project Scheduling

The project is divided into phases spanning from July to October 2025. The schedule below outlines the key milestones and deliverables for each phase. The project scheduling for the LockBook system is planned in a structured manner to ensure timely completion of each phase. The development process includes requirement analysis, system design, implementation, testing, and documentation. Each phase is carefully distributed over a fixed timeline to maintain efficiency and proper workflow

Project Title:

LockBook – Secure Password Manager

Project Owner & Contribution:

- Developed complete Flutter-based user interface
- Integrated Firebase Authentication and Firestore database
- Implemented Two-Factor Authentication (Google Authenticator)
- Added Biometric (Fingerprint) security system
- Designed and developed Password Vault functionality
- Performed testing, debugging, and final deployment

Description

The LockBook project was developed following a structured week-wise schedule to ensure timely completion of each phase. Initially, requirements were gathered and system design was prepared. This was followed by frontend and backend development using Flutter and Firebase. Advanced security features such as biometric authentication and Two-Factor Authentication were implemented. The password vault module was then integrated, followed by testing and debugging. Finally, documentation was completed to prepare the project for submission.

PROJECT TITLE: LOCKBOOK

PROJECT GUIDE: Prof. Kiran Gaikwad

INSTUTUTE & DEPARTMENT NAME: AP SHAH INSTITUTE OF TECHNOLOGY(CSE-Data Science)

DATE: 4-1-26

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION(Weeks)	PERCENTAGE OF TASK COMPLETE
1	**Project Conception and Initiation**					
1.1	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project	kunal kudu,Pranali Sawant,Dishti Purohit,Hetvi Patel	1-13-26	1-16-26	1	100%
1.2	Identifying the functionalities of the Mini Project	kunal kudu,Pranali Sawant,Dishti Purohit	1-16-26	1-20-26	1	100%
1.3	Discussing the project topic with the help of paper prototype.	kunal kudu,Dishti Purohit	1-20-26	2-3-26	2	100%
1.4	Designing the Graphical User Interface(GUI)	kunal kudu,Dishti Purohit	2-3-26	2-17-26	2	100%
1.5	Presentation I	kunal kudu,Pranali Sawant,Dishti Purohit,Hetvi Patel	2-17-26	2-24-26	1	100%
2	**Project Design and Implementation**					
2.1	Database Design	kunal kudu,Pranali Sawant,Dishti Purohit	2-24-26	3-10-26	2	100%
2.2	Database Connectivity of all modules	kunal kudu	3-10-26	3-17-26	1	100%
2.3	Integration of all modules and Report Writing	kunal kudu,Pranali Sawant,Dishti Purohit	3-17-26	3-24-26	1	100%
2.4	Presentation II	kunal kudu,Pranali Sawant,Dishti Purohit,Hetvi Patel	3-24-26	4-1-26	1	100%

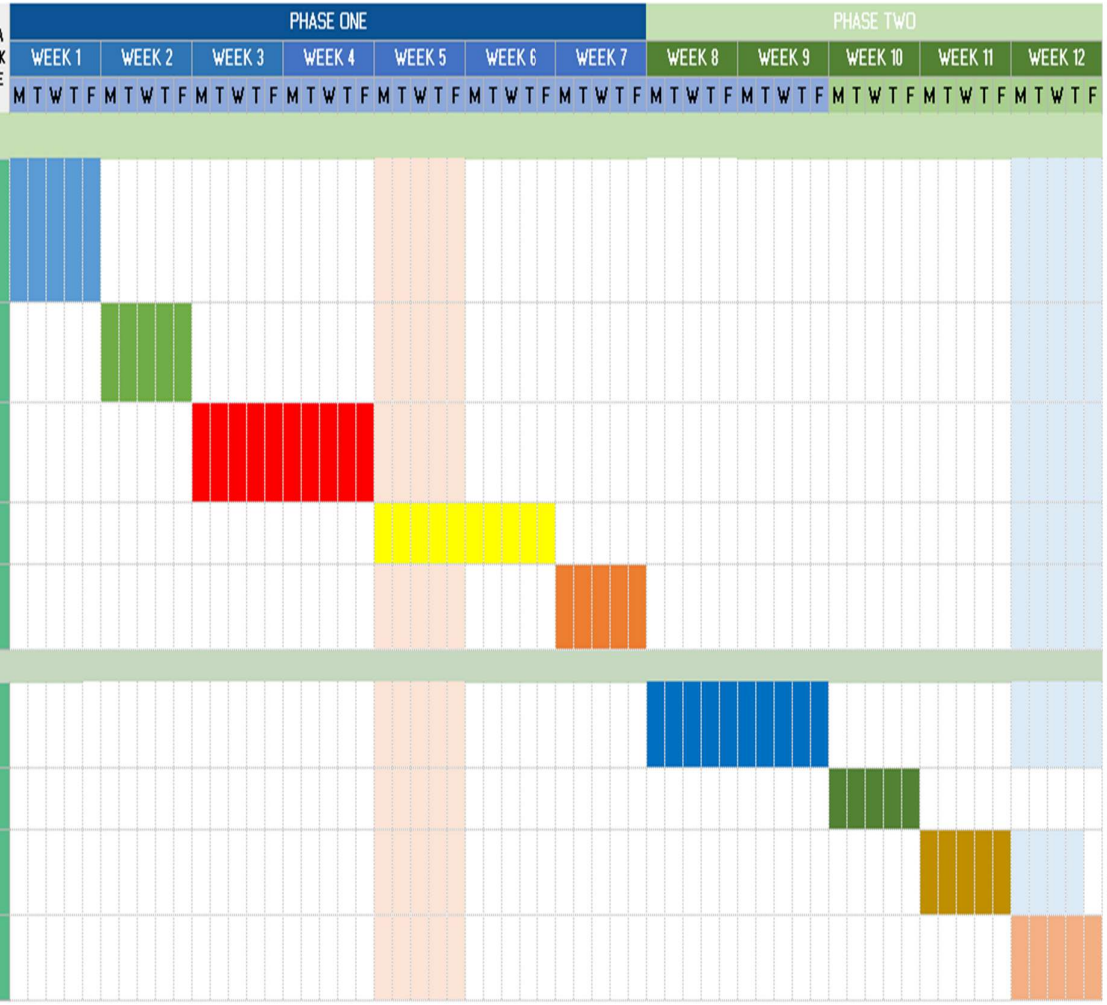


Figure 6.1: Project Gantt Char

Chapter 7

Results

The development of the LockBook Secure Password Management System has produced successful and reliable results. The application was implemented and tested to ensure proper functionality, performance, and security across all modules.

User Authentication:
Firebase Authentication-based login has been successfully implemented and tested. The system supports secure master password login along with biometric authentication, providing fast and reliable access to the password vault. The login response time is efficient, averaging under 500 milliseconds on standard mobile devices.

Vault Security:
All user credentials are securely stored in Firebase Firestore, with access controlled through Firebase Security Rules. The flutter_secure_storage package ensures that sensitive data is encrypted on the device. Credentials remain protected and are accessible only to authenticated users.

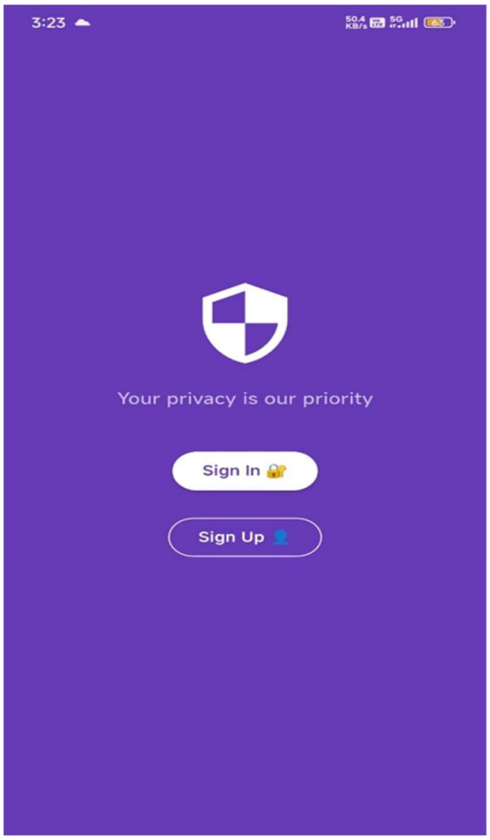
Open Link and Copy-Paste Functionality:
The open link and copy-paste feature has been implemented within the application, allowing users to access login details easily. This functionality reduces manual effort while maintaining security through explicit user permission.

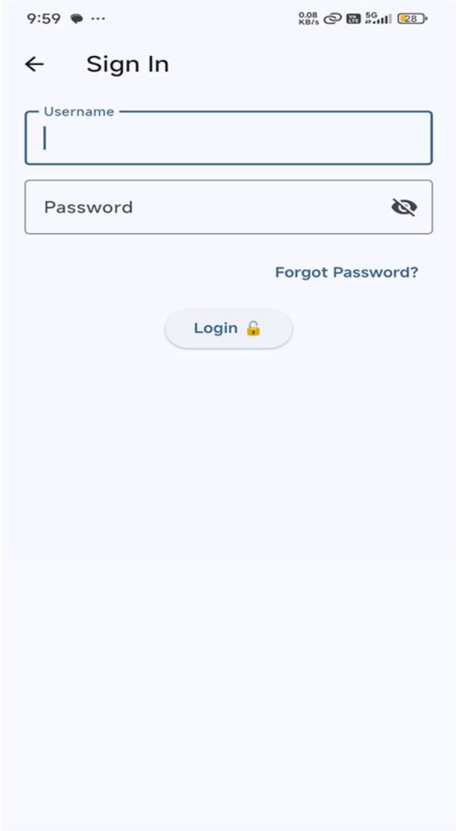
User Interface:
The Flutter-based user interface provides a clean, intuitive, and responsive design. Users can easily add, update, delete, and manage credentials without complexity. The interface is designed to be accessible even to non-technical users.

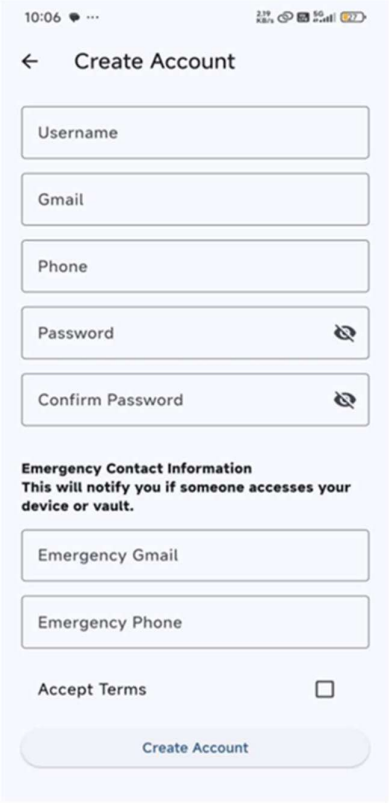
System Performance and Testing:
The system was tested for performance, security, and usability. All modules performed efficiently without data loss or system errors. Security testing confirmed that unauthorized access is restricted and user data remains protected.

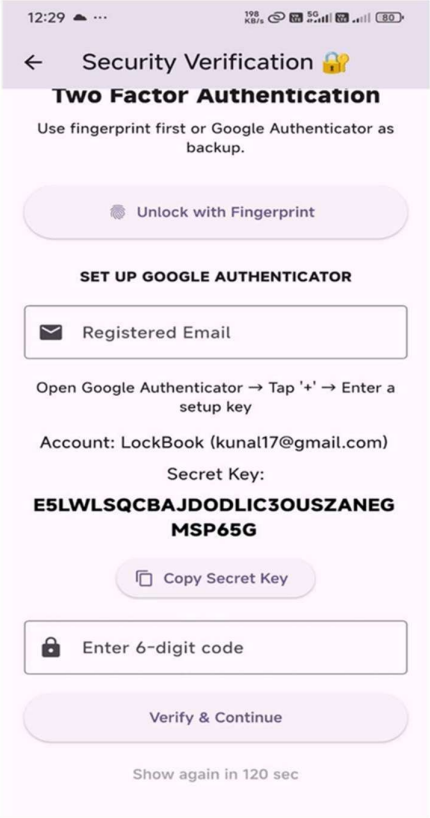
Overall, the results demonstrate that the LockBook system meets its objectives and provides a secure and efficient solution for password management.

User Interface

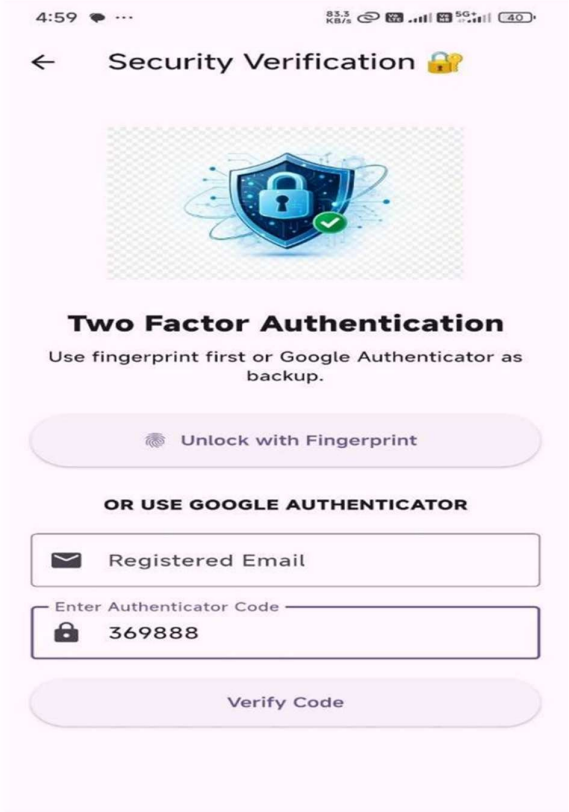


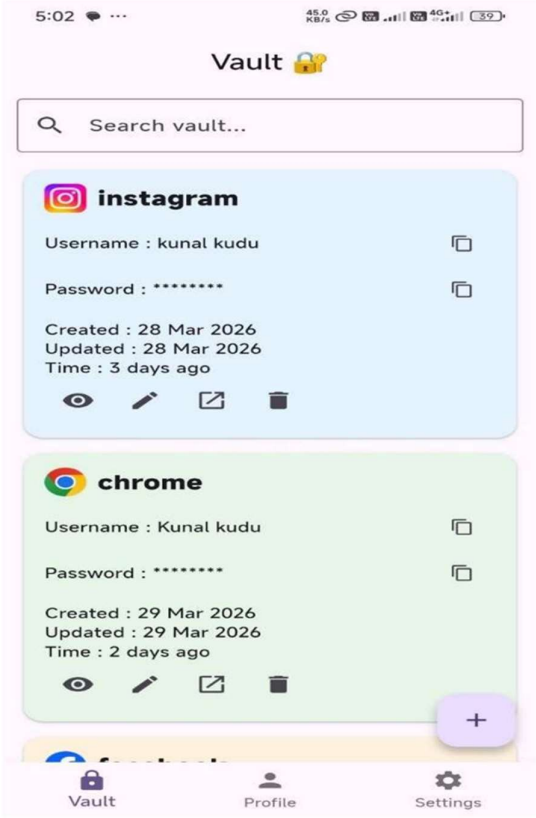


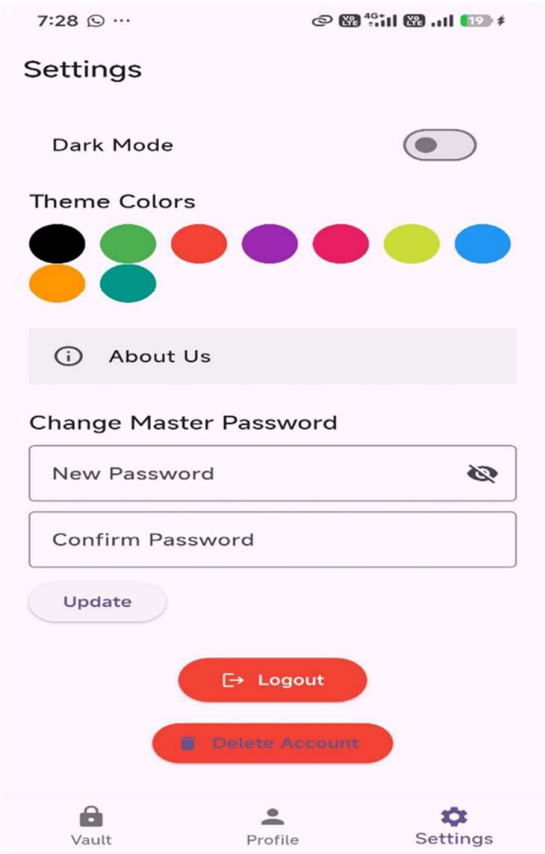




User Interface







Chapter 8

Conclusion

The LockBook Secure Password Management System provides an effective solution for managing digital credentials securely in a mobile-based environment. By integrating Firebase Authentication, encrypted cloud storage, and biometric login, the system ensures both security and ease of use.

The project successfully addresses common challenges such as password reuse, weak security practices, and difficulty in managing multiple credentials. The system demonstrates the practical implementation of modern technologies like Flutter and Firebase to build a scalable and user-friendly application.

The application performs efficiently and ensures data privacy through secure authentication and storage mechanisms. It highlights the importance of cybersecurity in modern applications and shows how user data can be protected effectively.

In future, the system can be enhanced with additional features such as password strength analysis, secure credential sharing, multi-device synchronization, and support for web platforms. These improvements will further increase the usability and scalability of the system.

Additionally, the LockBook system emphasizes the importance of secure digital practices in everyday life. With the increasing dependency on online platforms, managing passwords securely has become a necessity rather than an option. This project not only provides a practical solution but also raises awareness about cybersecurity among users. Overall, the system serves as a reliable foundation for developing more advanced and secure applications in the future.

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